

Level 1 Functional Diagram and N² Diagram — PD-RPM Remote Care System

A function is something the system must **do**, stated without naming the thing that does it, so that alternatives can be traded against the same requirement later. The six Level 1 functions below are the complete top-level decomposition of PD-RPM. The boundary is the one fixed earlier in the semester: the PD cycler, the EHR, and clinical decision authority sit outside it. Every flow is numbered; Table 1 gives its content and the N² diagram on page 2 places that same number in its interface cell.

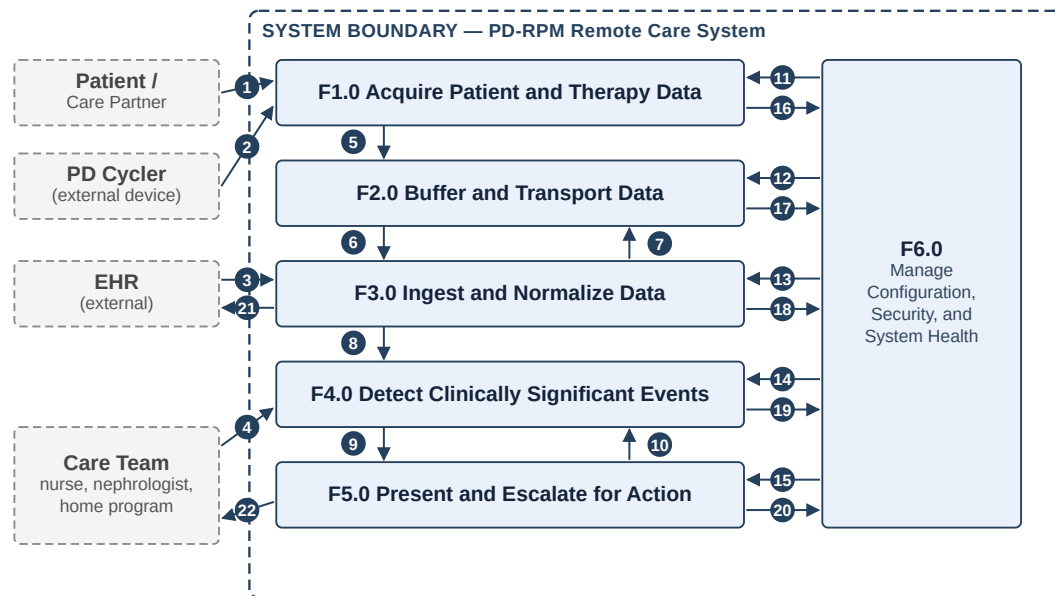


Figure 1. Level 1 functional diagram. Boxes inside the dashed boundary are functions the system performs; gray dashed boxes are external elements it interfaces with but does not perform. Numbered circles key to Table 1.

Table 1. Flow legend

1 Patient / Care Partner → F1.0	Weight, BP, symptom and exit-site entries	12 F6.0 → F2.0	Credentials, certificates, transport policy
2 PD Cycler (external) → F1.0	Volumes, dwell times, net UF, alarms	13 F6.0 → F3.0	Data model and mapping configuration
3 EHR (external) → F3.0	Demographics, prescription of record	14 F6.0 → F4.0	Threshold and detection rule parameters
4 Care Team → F4.0	Prescription changes and clinical annotations	15 F6.0 → F5.0	Notification routing and on-call roster
5 F1.0 → F2.0	Timestamped observation set	16 F1.0 → F6.0	Device health and connectivity status
6 F2.0 → F3.0	Encrypted upload, 30 min of session end	17 F2.0 → F6.0	Buffer depth and transfer latency
7 F3.0 → F2.0	Receipt acknowledgment; releases buffer	18 F3.0 → F6.0	Ingest completeness and rejected-record log
8 F3.0 → F4.0	Normalized, unit-consistent time series	19 F4.0 → F6.0	Rule execution and detection performance log
9 F4.0 → F5.0	Flagged event, severity, supporting trend	20 F5.0 → F6.0	Delivery confirmation and ack timing
10 F5.0 → F4.0	Clinician disposition: actionable or not	21 F3.0 → EHR (external)	Treatment summaries written back
11 F6.0 → F1.0	Device configuration and acquisition schedule	22 F5.0 → Care Team	Triage queue views, alerts, and pages

N² Diagram — PD-RPM Remote Care System

The N² diagram below carries the same six functions on its diagonal. **Outputs leave a function horizontally along its row; inputs enter a function vertically down its column**, so the cell at row F3.0 and column F4.0 holds what F3.0 delivers to F4.0. A blank cell means no interface exists between that ordered pair. External interfaces are carried in the flanking columns rather than omitted, since the boundary crossings are where this system is most likely to fail.

Inputs from outside the boundary	F1.0	F2.0	F3.0	F4.0	F5.0	F6.0	Outputs beyond the boundary
1 vitals, symptom and exit-site entries (Patient / Care Partner) 2 treatment records (PD Cycler)	F1.0 Acquire Patient and Therapy Data	5 observation set				16 device health status	
		F2.0 Buffer and Transport Data	6 encrypted upload			17 buffer depth, latency	
3 demographics, prescription of record (EHR)		7 receipt acknowledgment	F3.0 Ingest and Normalize Data	8 normalized time series		18 ingest completeness	21 treatment summaries (EHR)
4 prescription changes, clinical annotations (Care Team)				F4.0 Detect Clinically Significant Events	9 flagged event + severity	19 rule execution log	
				10 clinician disposition	F5.0 Present and Escalate for Action	20 delivery confirmation	22 queue views, alerts, pages (Care Team)
	11 device configuration	12 credentials, policy	13 mapping configuration	14 threshold parameters	15 routing, on-call roster	F6.0 Manage Configuration, Security, System Health	

Alignment check. All twenty-two numbered flows in Figure 1 appear exactly once in the matrix: flows 1–4 in the left column, 5–20 in the sixteen populated off-diagonal cells, and 21–22 in the right column. Fourteen of the thirty off-diagonal pairs are empty, which is itself a result — it says acquisition never talks to detection directly, and that every path to a clinician passes through F5.0.